

SRISHTI RAO

Software Engineer

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SUMMARY

An aspiring software engineer with experience in developing secure backend services, microservices, and Financial Technologies (FinTech) applications using Java, Spring Boot, and Apache Kafka. Currently pursuing a B.Tech in Artificial Intelligence, with practical exposure to AI/ML projects and modern web technologies.

EXPERIENCE

WOW Finstack

Software Developer (Intern)

Hybrid

Dec 2024 - Present

- Developed secure **RESTful backend services** and APIs with Java, Spring Boot, implementing best practices for authentication, authorization, and error handling.
- Designed and implemented a scalable microservice for **dynamic QR code generation** with support for customization and integration between services.
- Implemented the **User Management module** in a **Card Management System** to support onboarding, authentication, and account profile management.
- Conducted extensive **testing and debugging of the Issuer System** using Visa Simulator to ensure reliable card transaction processing and payment management.

Sardar Vallabhbhai National Institute of Technology

Research Intern

Onsite

Aug 2025 - Dec 2025

- Developed a **Multi-Resolution Ensemble (MRE)** framework using 9 lightweight CNNs (EfficientNet, MobileNet, ShuffleNet) to classify breast cancer histopathology images.
- Engineered **MRPE**, a novel weighting scheme that optimizes the balance between model accuracy and inference speed for clinical use.
- Achieved a **95.00% test accuracy** with an EfficientNet-B0 backbone, significantly surpassing the 87% ICIAR 2018 BACH challenge benchmark.
- Implemented a **quantitative XAI framework** using Grad-CAM and IoU to measure spatial overlap between AI heatmaps and pathologist regions of interest.
- Established a statistically significant **positive correlation (p=0.02)** between AI activations and human-centric diagnostic scores, improving model transparency.
- Benchmarked **FP16 vs. FP32 inference** on Apple M4 hardware, identifying critical performance bottlenecks in specialized hardware kernels.
- Built a **Gradio web interface** to provide a “glass-box” diagnostic tool for real-time image analysis and uncertainty visualization.

PERSONAL PROJECTS

- **Breast Cancer Classification from Histopathology Images**

The project takes a medical image and classifies the type of breast cancer using ViT and ResNet models. Provides a side-by-side visual explanation using Grad-CAM and SHAP to make the predictions interpretable, regulated by the XAI regulator for reliable results.

- **Voice Assistant for Elderly Care**

A voice-enabled assistant designed to support elderly users with reminders, weather information, and emergency alerts providing an intuitive and accessible interface to enhance independence and safety.

- **Dynamic QR Management**

Designed and implemented a system to generate user-specific dynamic QR codes with optional image customization.

SKILLS

- **Programming Languages:** Java, Python, C, C++, JavaScript, HTML, CSS
- **Frameworks & Technologies:** Spring Boot, React.js, Vite, TailwindCSS, PyTorch, Docker, Apache Kafka
- **Databases:** PostgreSQL, MySQL (Database Design, Query Optimization, SQL)
- **AI & Machine Learning:** Model Training, Data Preprocessing, Supervised/Unsupervised Learning, Python Libraries (NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV)
- **Security:** JWT, HMAC, Encryption
- **Domain Knowledge:** Financial Technologies (FinTech), Card Transaction Processing, Issuer System, VTS (Visa Simulator)
- **Soft Skills:** Leadership, Adaptability, Collaboration, Self-Learning, Communication, Problem Solving

EDUCATION

Sardar Vallabhbhai National Institute of Technology
Bachelor of Technology in Artificial Intelligence

Surat, Gujarat
2023 - Present

- **CGPA:** 6.95/10

Global Indian International School
Secondary and Senior Secondary Education

Singapore
2017 - 2023

- **Class 10:** 95.8%
- **Class 12:** 89%